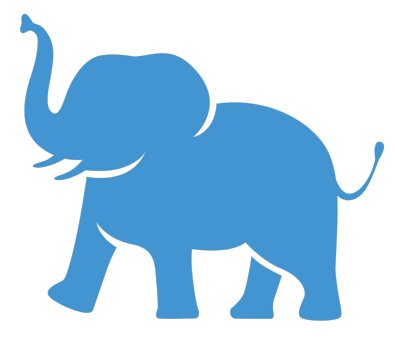
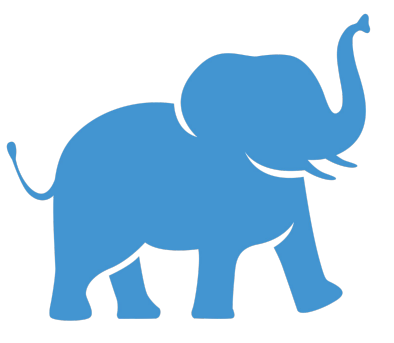




An offline approach to fNIRS-guided reinforcement learning for robot behavior



Julia Santaniello

julia.santaniello@tufts.edu

Madelaine Brower

madelaine.brower@tufts.edu

Benson Jiang

benson.jiang@tufts.edu

Donatello Sassaroli

donatello.sassaroli@tufts.edu

Chenyuan Zhang

czhang22@tufts.edu

Robert Jacob

jacob@cs.tufts.edu

Jivko Sinapov

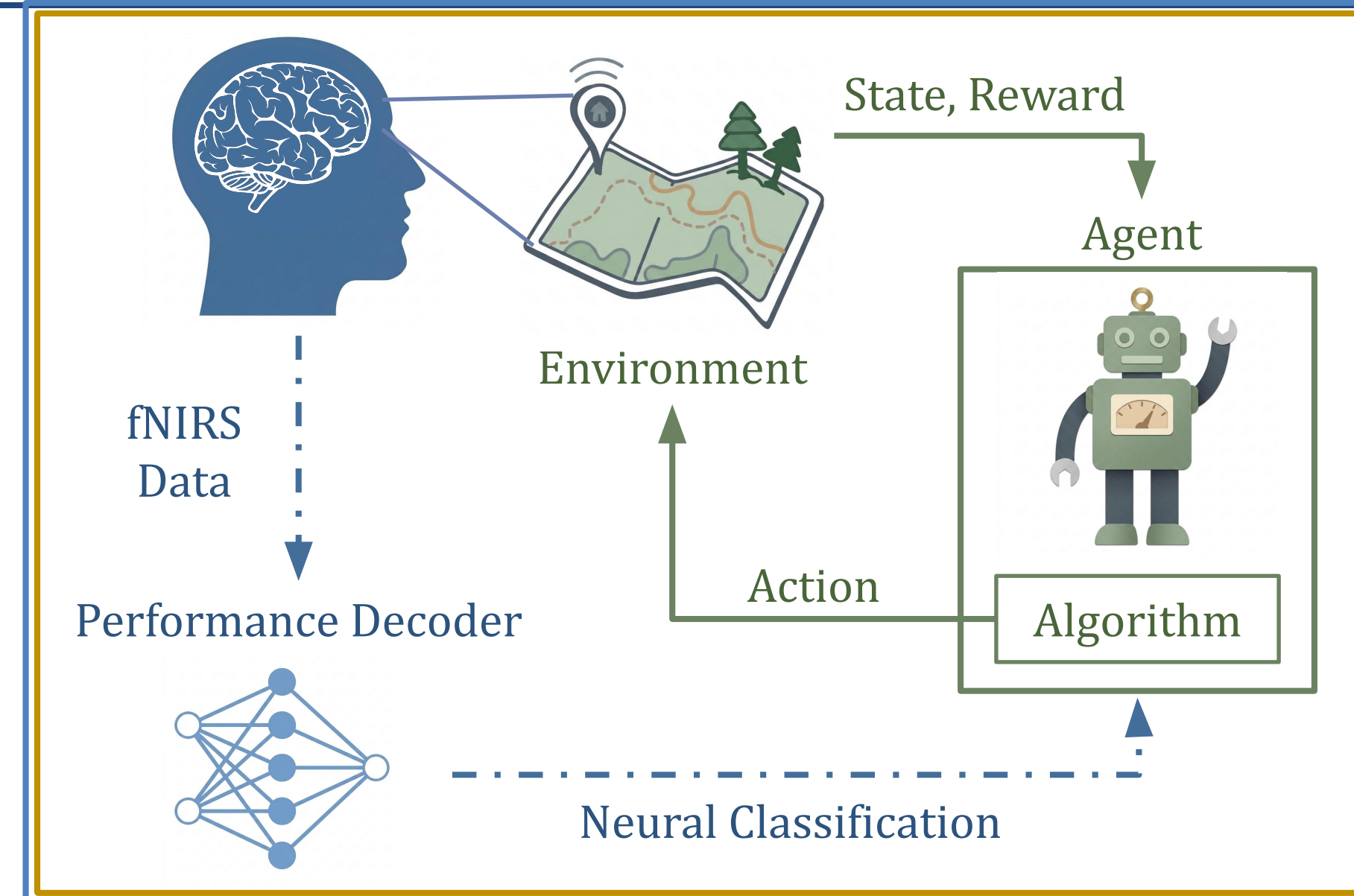
jivko.sinapov@tufts.edu

Introduction

- **Human-in-the-Loop Reinforcement Learning (HITL-RL)** enhances agent training through using feedback from human “teachers”
- Most algorithms rely on preference or demonstration data as evaluative feedback. Feedback acquisition often **increases user workload** due to sustained attention, decision making and/or active demonstration
- **Passive Brain-Computer Interfaces (BCI)** assess user cognitive states through various neuroimaging devices
- **Functional Near-Infrared Spectroscopy (fNIRS)** is one such device that measures the change in hemodynamic response in the brain

Questions:

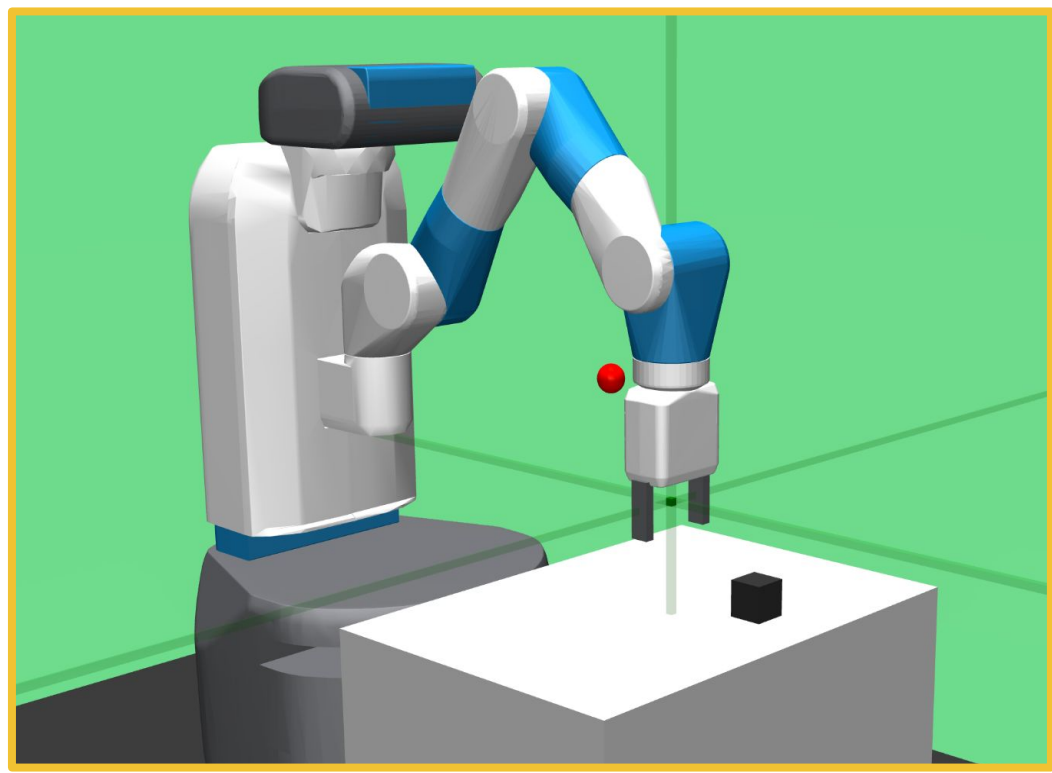
1. Can we improve robot learning by applying passive brain signals to a reinforcement learning training loop?
2. How does decoder noise influence learning?



Setup



A Participant sits at a computer and **observes** or **guides** an agent during a task. fNIRS estimates prefrontal cortex **hemodynamic activity**.



Robot Fetch and Place:

The robot's goal is to place the black cube in the position indicated by the red sphere.

Machine Learning

Time Series Classification:

Preliminary work shows that fNIRS data can be mapped to **robot performance**. We frame this as a time series classification problem, and predict agent performance from windows of neural feature vectors.

Discrete Labels: As the user observes the agent, we force the agent to take **optimal, sub-optimal or worst-case actions**. These classes are used as labels to train a **classifier** that can predict agent performance from fNIRS signals alone.

Continuous Labels (Multi-Policy Agreement):

We applied **Euclidean Distance** to calculate the error between the agent's chosen action and the chosen actions of ten (10) near-optimal policies. These ten scores were averaged and used as a continuous label for **regression analyses**.

The core RL algorithm runs normally, except:

- Sparse Environmental Rewards
- TD-error Priorities
- Q-Target Values

may be augmented by the neural signal.

Ablations:

- Beta (Scaling Factor)
 - $\beta \in (0.5, 1.0, 5.0)$
 - To test if neural signal shaping benefits from a scaling factor
- Decoder Noise:
 - Noise $\in (0.0, 0.5, 1.0)$
 - To test if the neural signal is the main driver of performance gains and to characterize how much noise the framework can withstand before gains disappear
- Finetune Checkpoints
 - Checkpoint $\in (0.0, 0.2, 0.5, 0.7)$
 - To test whether the framework is best applied in early, mid or late-stage training

Algorithm 1: Neural optimality shaping.

Input: reward r , td-error priority δ , q-aug value q , confidence c , target class $\hat{y}$, flags $\mathcal{F}$, scale β , expected values μ

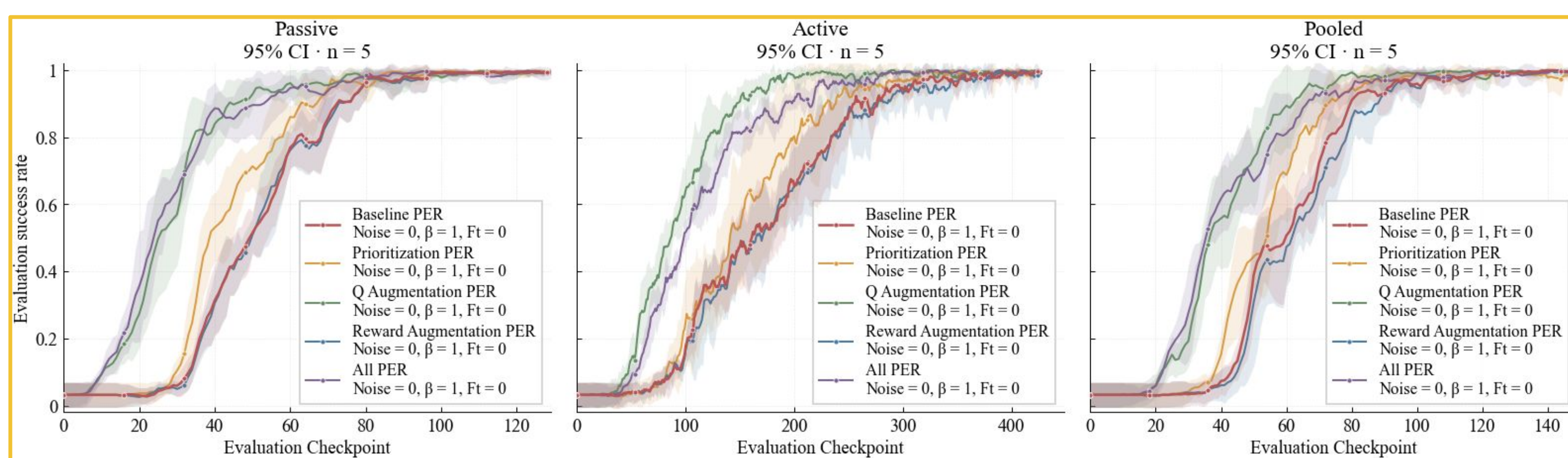
Output: r_t, δ_t, q_t

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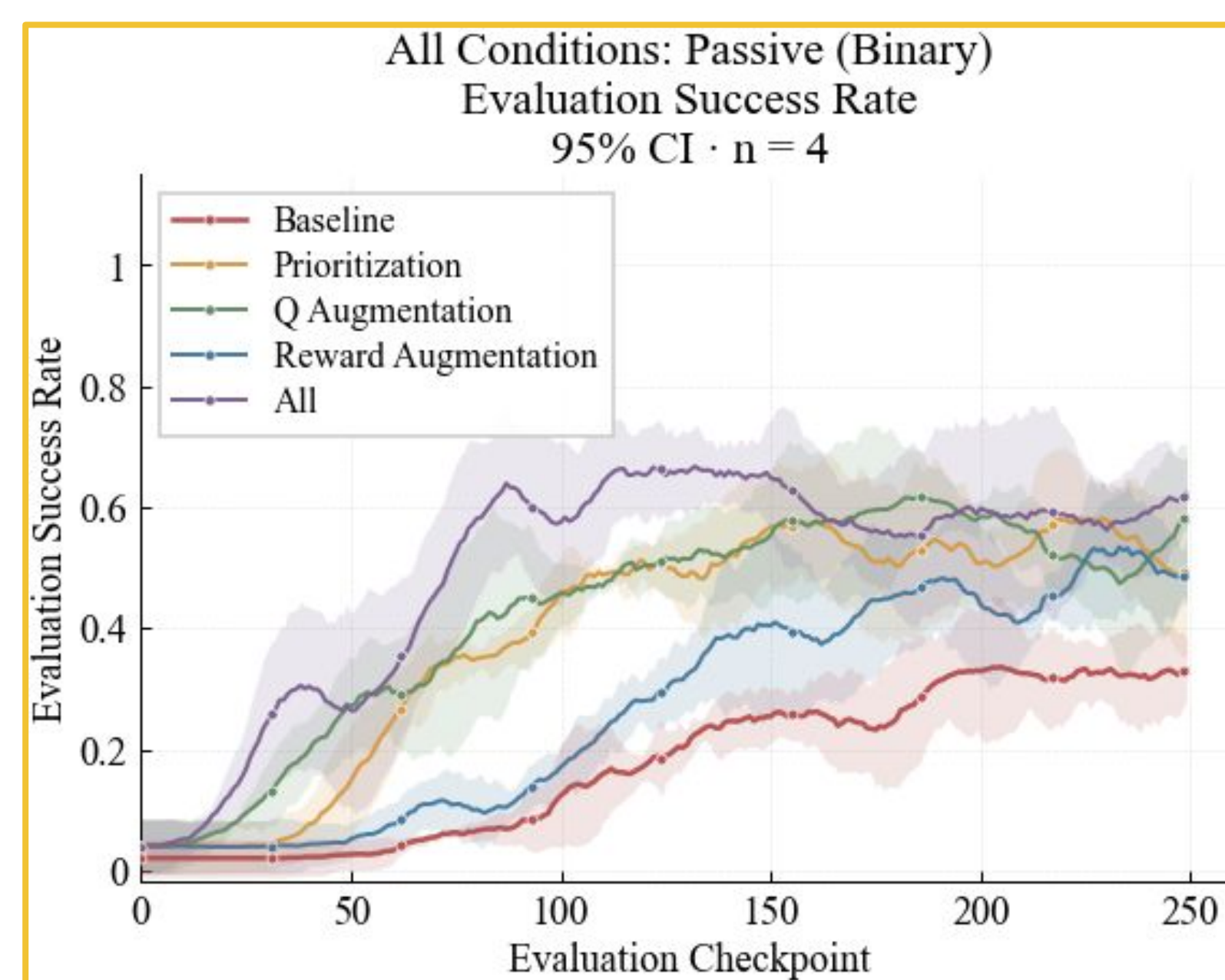
1:  $r_t \leftarrow r; \delta_t \leftarrow \delta; q_t \leftarrow q$ 
2: if  $1 \in \mathcal{F}$  then
3:    $r_t \leftarrow \beta(r_t + c \cdot \mu_r[\hat{y}])$   $\triangleright$  reward augmentation
4: end if
5: if  $2 \in \mathcal{F}$  then
6:    $\delta_t \leftarrow \beta(\delta_t + c \cdot \mu_\delta[\hat{y}])$   $\triangleright$  priority augmentation
7: end if
8: if  $3 \in \mathcal{F}$  then
9:    $q_t \leftarrow \beta(q_t + c \cdot \mu_q[\hat{y}])$   $\triangleright$  Q-augmentation
10: end if
11: return  $r_t, \delta_t, q_t$ 

```

Results

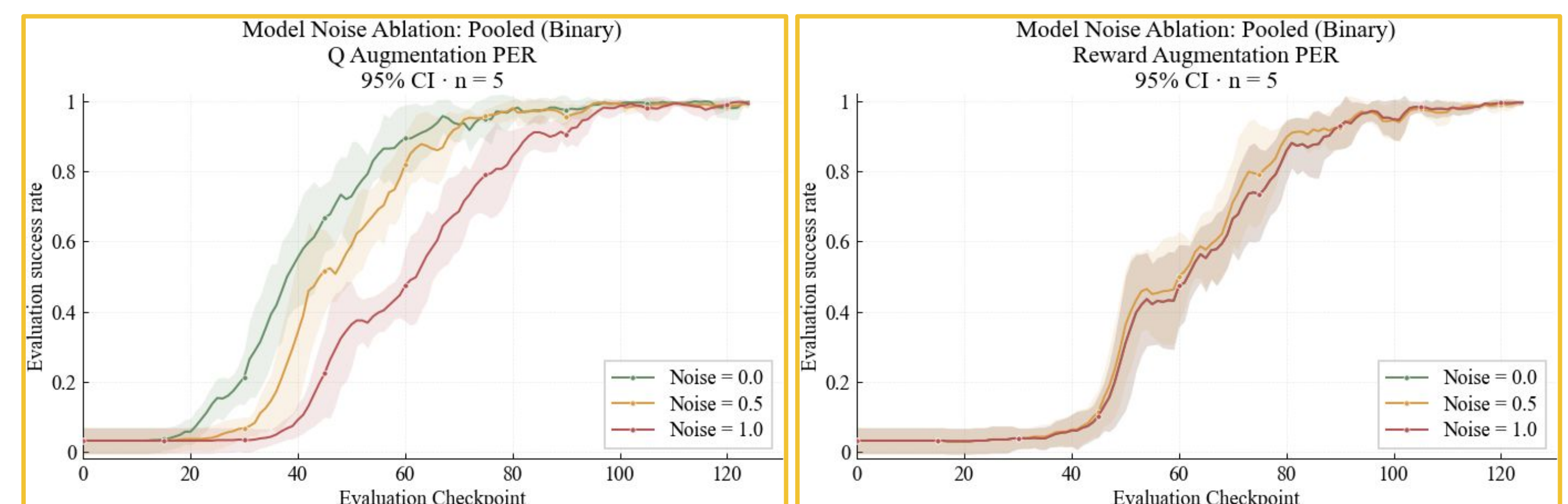


Dense Reward Functions: We further tested this framework on a robot learning from a dense reward function. All conditions outperformed baseline in this experimental condition.



Results Across Interaction Types: Binary experiments show that robot learning is accelerated by applying the neural signal to augment Q-Targets and Priorities.

Robots trained on Passive tasks trained faster than robots trained on Active tasks or a mixture of the two (Pooled).



Decoder Noise on Robot Performance: Noise ablations show that for conditions with significant gains, adding more noise degrades performance back towards baseline. For conditions without significant gains, there is no effect. This implies that performance gains are a direct result of the neural signal.